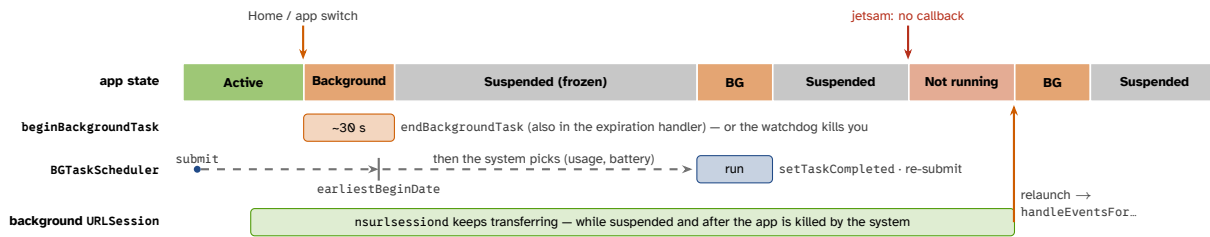


In one line: After the user leaves, an app gets a few seconds and is then **suspended**: in memory, no CPU. Every way of running later is either a short **grant** (`beginBackgroundTask`), a **request** the system schedules when *it* chooses (`BGTaskScheduler`, silent push), a **hand-off** to a daemon (background URLSession), or a **declared mode** that holds only while it is really in use (audio, location, ...). None of them is guaranteed.



How it works

- **States:** Not running · Inactive · Active · **Background** (running, off screen) · **Suspended** (in memory, threads frozen). A suspended app can be purged with **no** callback, so save state on entering the background.
- **`beginBackgroundTask`:** time to *finish* a save or upload, ~30 s (`backgroundTimeRemaining` is an estimate). Every begin needs an `endBackgroundTask`, also in the expiration handler.
- **`BGTaskScheduler`** (iOS 13): **`BGAppRefreshTask`** = short (~30 s), keeps content fresh; **`BGProcessingTask`** = minutes, deferrable maintenance (DB clean-up, ML), usually idle/overnight, can set `requiresExternalPower` / `requiresNetworkConnectivity`. Needs the identifiers in `Info.plist BGTaskSchedulerPermittedIdentifiers`, background mode fetch and/or processing, and register for each **before launch finishes** (iOS may launch the app *only* to run it). Tasks are **one-shot**, so re-submit. The system decides when, from app usage, battery, Low Power Mode and the *Background App Refresh* switch.
- **`BGContinuedProcessingTask`** (iOS 26): *user-started* work (export) that **starts in the foreground** and goes on in the background; `ProgressReporting` shown in a cancellable **Live Activity**; little progress = terminated first; GPU needs an entitlement.
- **Background URLSession:** `.background(withIdentifier:)` config, delegate-based (no completion handlers), uploads from a file. The daemon transfers, then relaunches the app in `handleEventsForBackgroundURLSession`: store its handler, recreate the session with the **same id**, call the handler on main after `urlSessionDidFinishEvents`. `isDiscretionary` lets iOS wait for Wi-Fi and power.
- **Background modes** (`UIBackgroundModes`): audio, location, voip (PushKit + CallKit), bluetooth-central/-peripheral, remote-notification, external-accessory, fetch, processing. The continuous ones hold only *while doing that work*. **Silent push** (`content-available:1`, priority 5) = a throttled server-side wake for ~30 s.

Which tool for which job

finish a save on leaving	<code>beginBackgroundTask</code>
fresh content at next open	<code>BGAppRefreshTask</code>
nightly DB / ML clean-up	<code>BGProcessingTask</code>
user-started export	<code>BGContinuedProcessingTask</code> (26)
download 1 GB	background URLSession
server says "new data"	silent push (throttled)
playing, navigating, calling	the matching background mode

What gets you killed

- **Jetsam:** memory pressure; background/suspended and big apps go first, with no callback. Keep the background footprint small.
- **Watchdog** (`0x8badf00d`): main thread blocked too long (slow launch, sync I/O), or a background task never ended.
- **`0xdead10cc`:** suspended while holding a file lock (SQLite in an App Group). Wrap the write in a background task.
- Too much background CPU, or a faked mode (silent audio).

Example

```
let bg = BGTaskScheduler.shared, rid = "com.you.refresh"
// in didFinishLaunching, BEFORE it returns:
bg.register(forTaskWithIdentifier: rid, using: nil) {
    handle($0 as! BGAppRefreshTask) }
func schedule() {
    let r = BGAppRefreshTaskRequest(identifier: rid)
    r.earliestBeginDate = .now + 15 * 60 // not before; no promise
    try? bg.submit(r) }
func handle(_ task: BGAppRefreshTask) {
    schedule() // one-shot: re-arm first
    let t = Task { task.setTaskCompleted(success: await sync()) }
    task.expirationHandler = { t.cancel() } }
var id = UIBackgroundTaskIdentifier.invalid // finish a save
id = app.beginBackgroundTask { app.endBackgroundTask(id) }
store.save { app.endBackgroundTask(id) }
```

Debug (the debugger stops tasks running): pause, then in LLDB

```
e -l objc -- (void)[[BGTaskScheduler sharedScheduler]
    _simulateLaunchForTaskWithIdentifier:@"com.you.refresh"]
(_simulateExpiration... tests the expiration handler).
```

Traps

- ! "My BG task never runs": running under the debugger, never re-submitted, identifier missing from `Info.plist`, registered too late, user rarely opens the app, Low Power Mode, or Background App Refresh is off.
- ! `earliestBeginDate` is a *not-before* date, not a schedule.
- ! After the user **force-quits** (swipe up): no BG tasks, no silent push, and pending background-session transfers are **cancelled**. VoIP PushKit still works.
- ! On a background-session relaunch, recreate the session with the *same identifier* or its delegate events are never delivered.
- ! Skipping `setTaskCompleted` or the URLSession handler costs future background time.
- ! `beginBackgroundTask` is for *finishing*, not polling (was ~3 min, now ~30 s).

Remember

Grant · Request · Hand-off · Mode — none is a promise.